

ZoomPursuit: **Smooth Pursuit Gaze Interaction** **for General-Purpose GUIs**

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Gaze Interaction

Accessibility

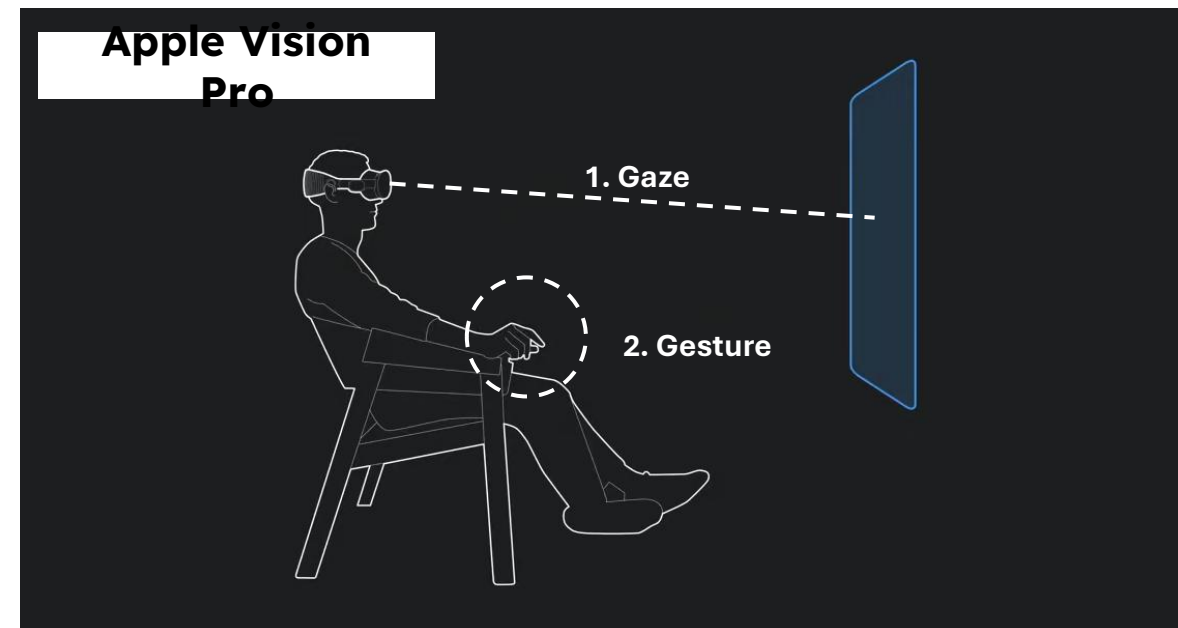
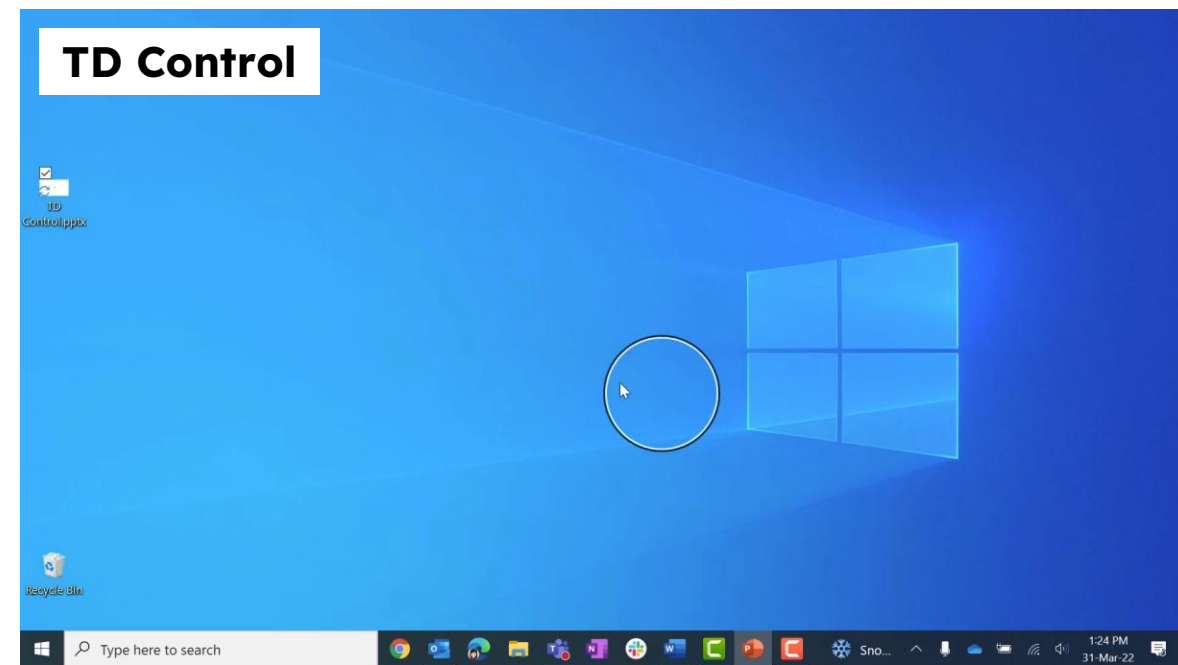
Promising Input modality in MR

- Fast and intuitive

Problems:

- Inaccuracy (AVP: 0.7° , right after calibration)
- Need explicit calibration before use
- Calibration drift overtime (AVP: 1.15° , re-wear it)

One approach to solve the problems is Smooth Pursuit



Smooth Pursuit

How it works:

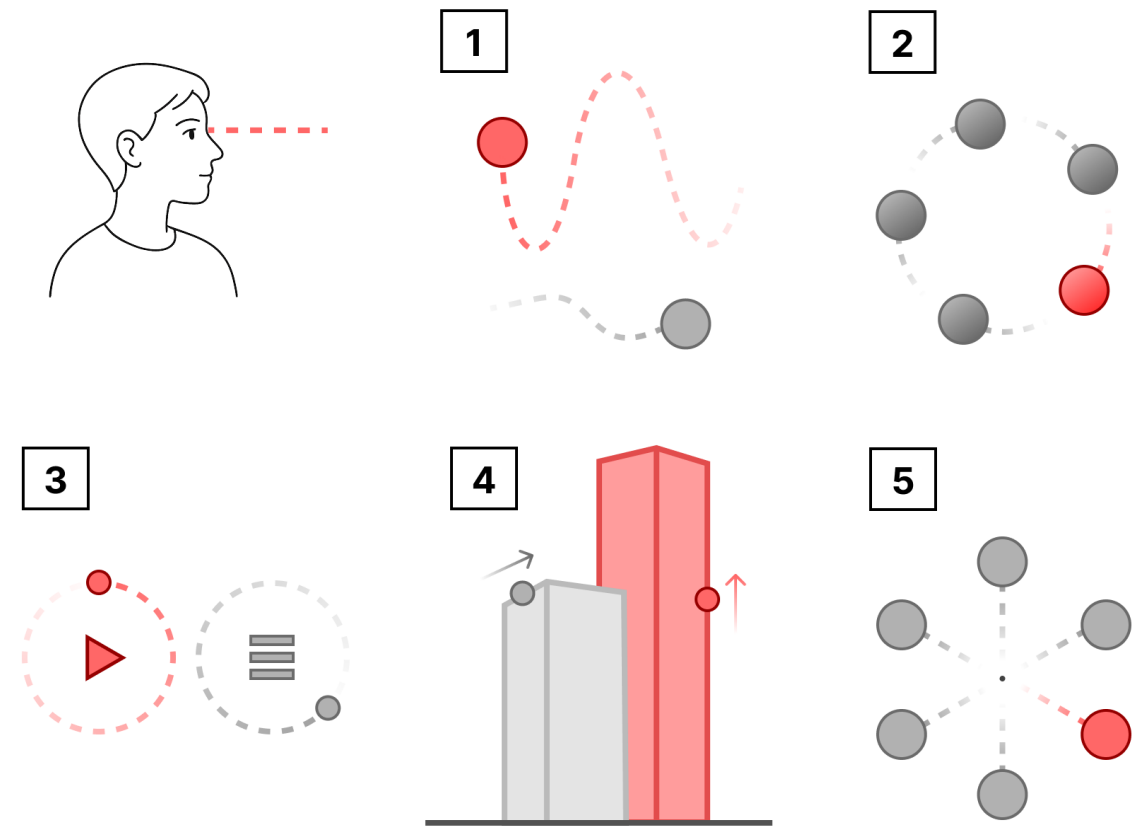
- Human eyes can easily follow a smooth motion
- Select by matching gaze & object motion

Benefits:

- Robust to absolute error / explicit-calibration-free
- Accurate (can select among hundreds of objects)
- Dynamic on-the-fly calibration

But it can't support general-purpose GUIs:

- Requires GUI access to inject motion stimuli
- Can't interact with arbitrary point (e.g., pickup color of a pixel)

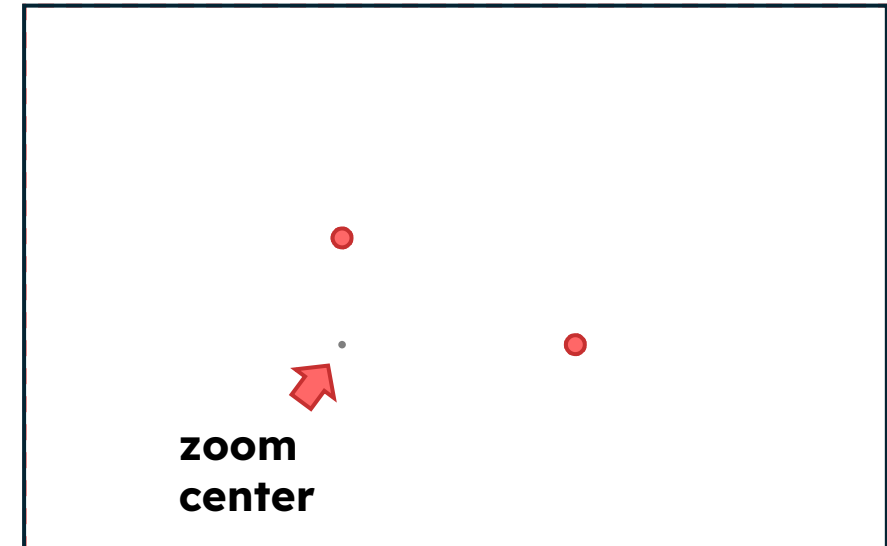


► **Overall research question:**
*How can we design a smooth pursuit technique that **retains its benefits** and is **applicable** to general-purpose GUIs?*

ZoomPursuit

Main idea: Zoom in creates unique motion vectors for every pixel on the screen from zoom center

Illustration

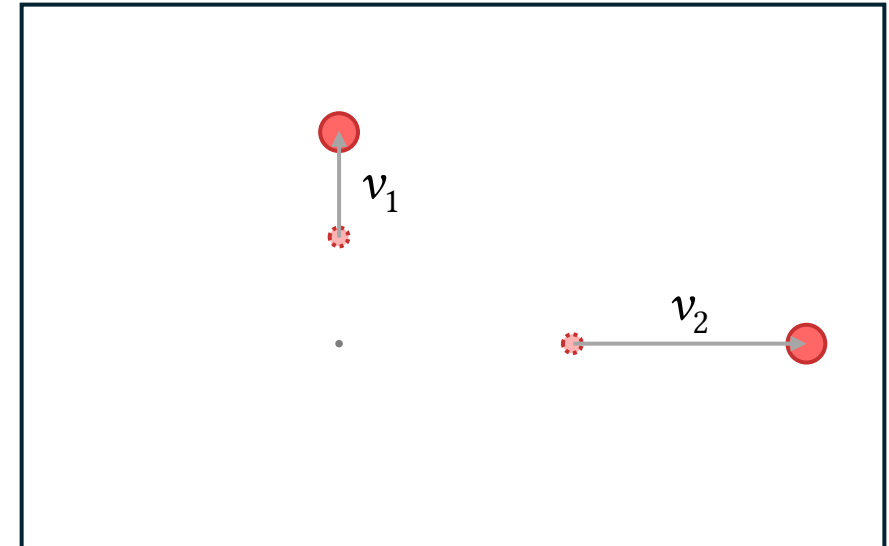


ZoomPursuit

How does ZoomPursuit work?

- We use the measured gaze as zoom center to minimize pursuit distance
- In theory, we can find out the motion vector that fit the best to the gaze movement.
- In practice: calculate correction vector

Illustration





ZoomPursuit

How does ZoomPursuit work?

- We use the measured gaze as zoom center to minimize pursuit distance
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- In practice: calculate correction vector

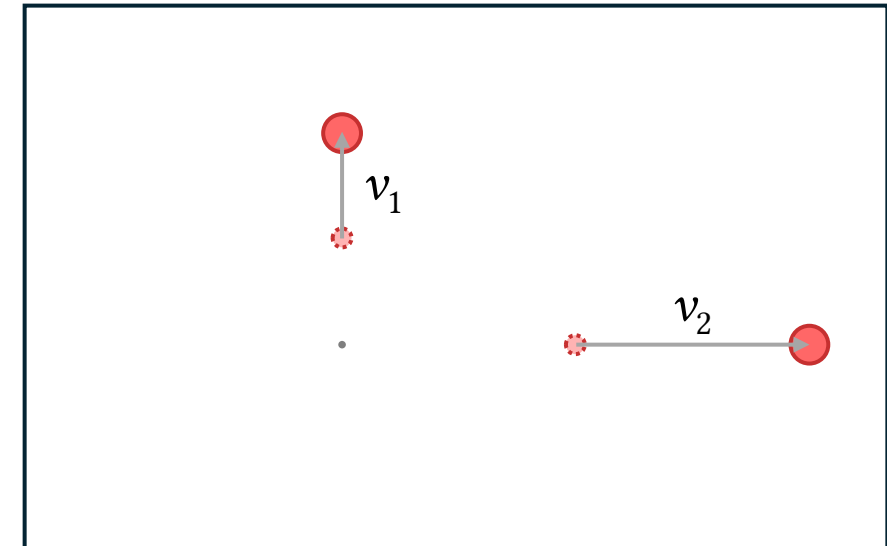
Applicable to general-purpose GUIs :

- Zoom in is a system-level API.
- Every point has a unique motion vector

► Overall research question:

*How can we design a smooth pursuit technique that **retains its benefits(need evaluation)** and **is applicable (✓)** to general-purpose GUIs?*

Illustration



Baseline Evaluation Study

RQ1: *How do ZoomPursuit and two baseline techniques compare in **performance** and **user experience** under **accurate** and **drifted** calibration conditions?*

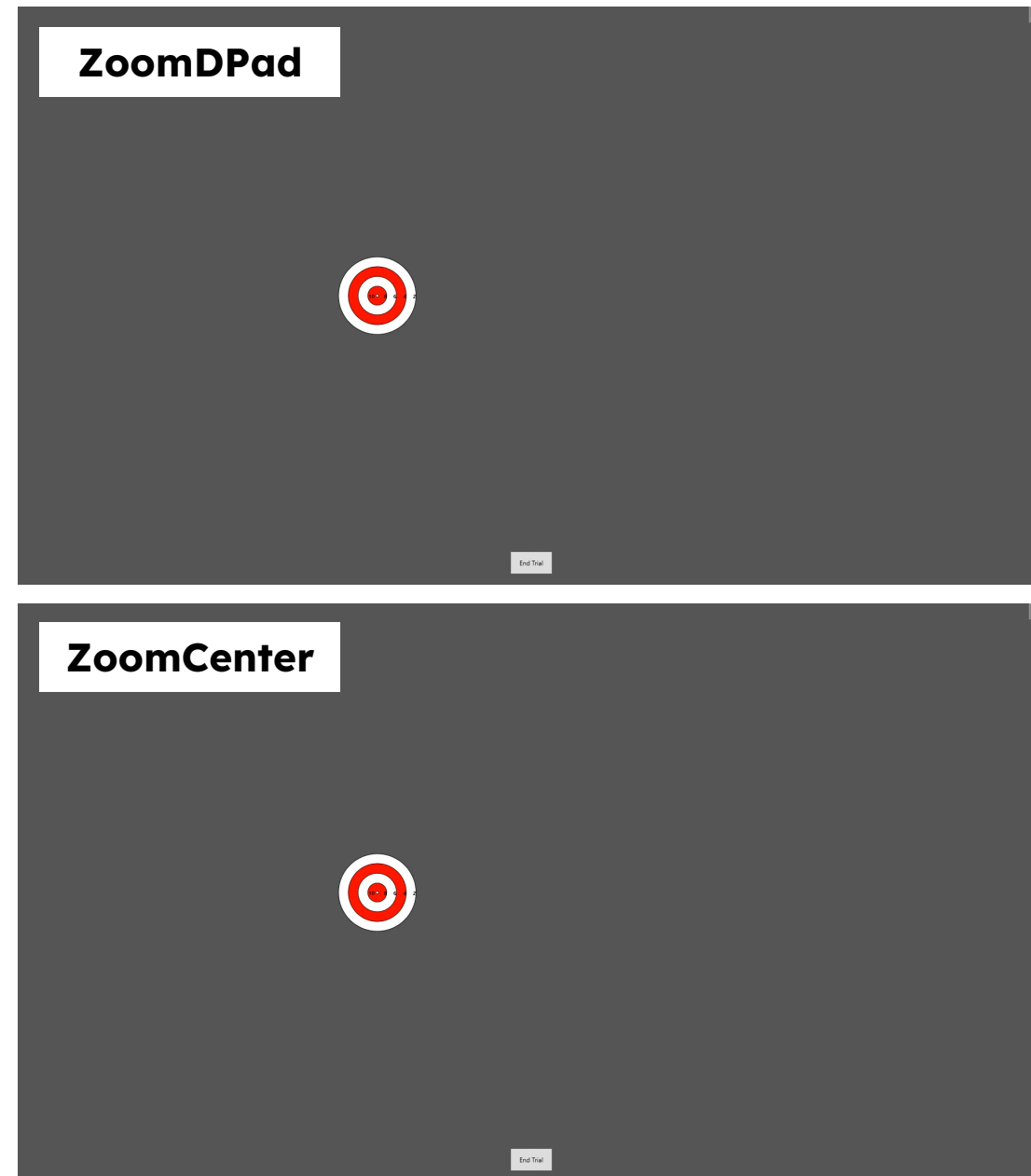
Mixed design:

(between-subjects) 2 calibration levels (1° vs. 3°)

(within-subjects) 3 techniques

- ZoomPursuit
- ZoomDPad (gaze-controlled D-pad component)
- ZoomCenter (magnify on the center)

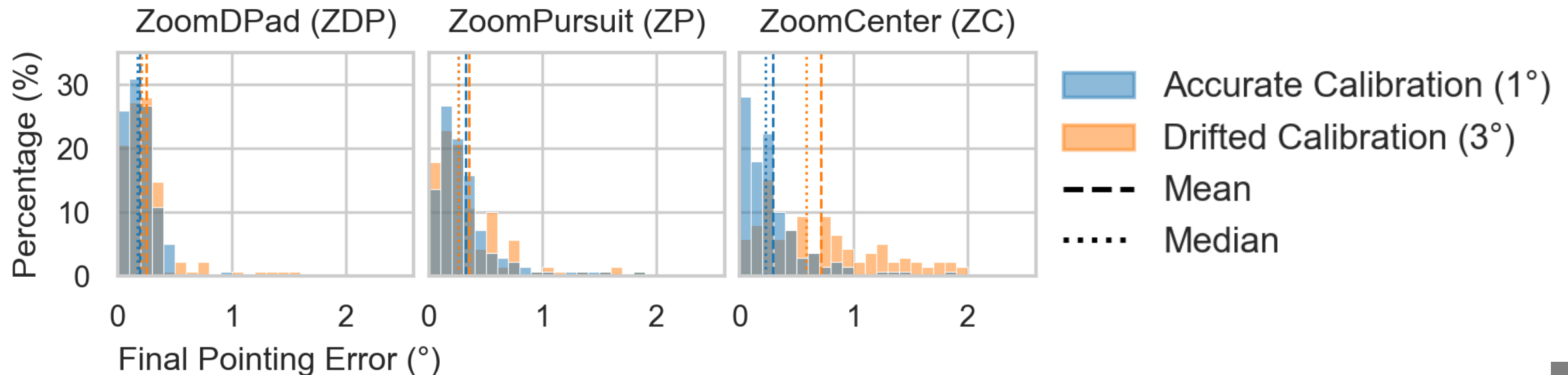
Task for each technique: 20 targets selection



Result - Accuracy

Main method: Linear Mixed Models (LMMs)

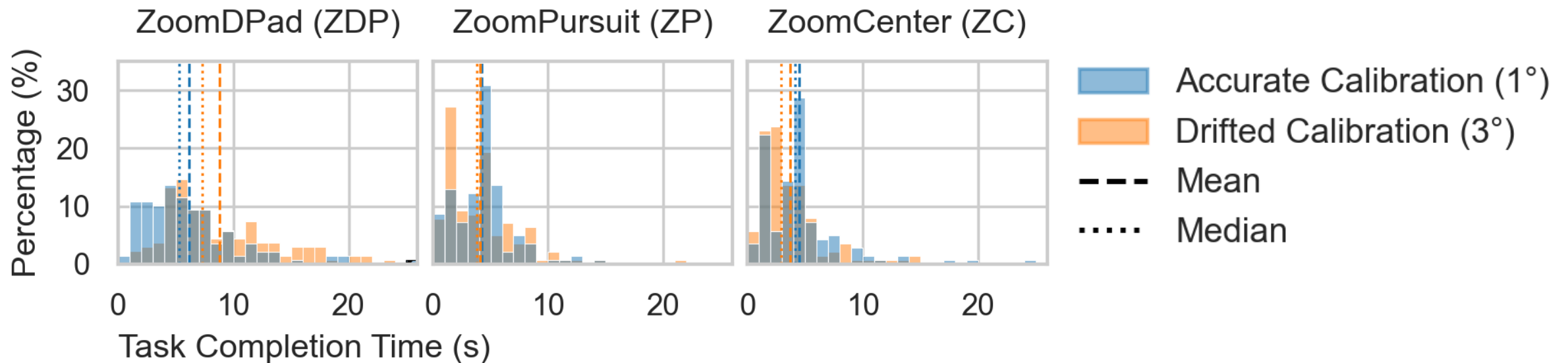
- **Under drifted condition:**
ZoomPursuit outperform ZoomCenter, no significant compared to ZoomDPad
 - **Under accurate condition:**
All accurate ($< 0.5^\circ$), ZoomDPad is the best
- ↓ **Visualization:** Distributions of final pointing error (280 trials per technique)



Result - Speed

Main method: Linear Mixed Models (LMMs)

- **ZoomDPad** is slowest; under drifted condition **90% longer than ZoomPursuit**
 - ZoomPursuit is similar to ZoomCenter across both calibrations
- ↓ **Visualization:** Distributions of task time (280 trials per technique)



Workload & Usability

Workload (NASA-TLX):

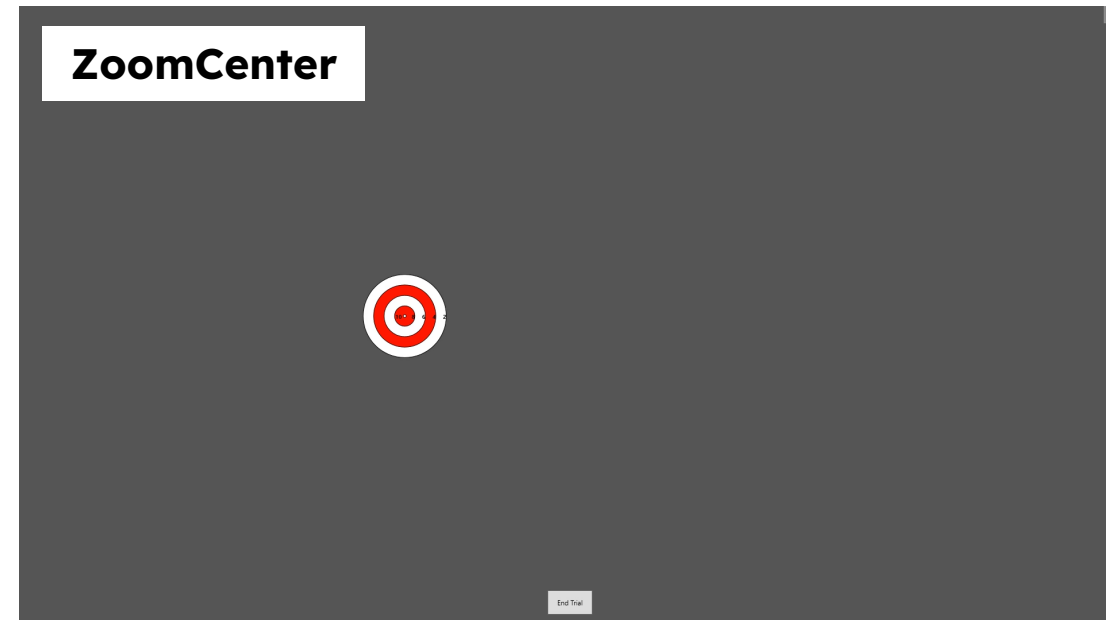
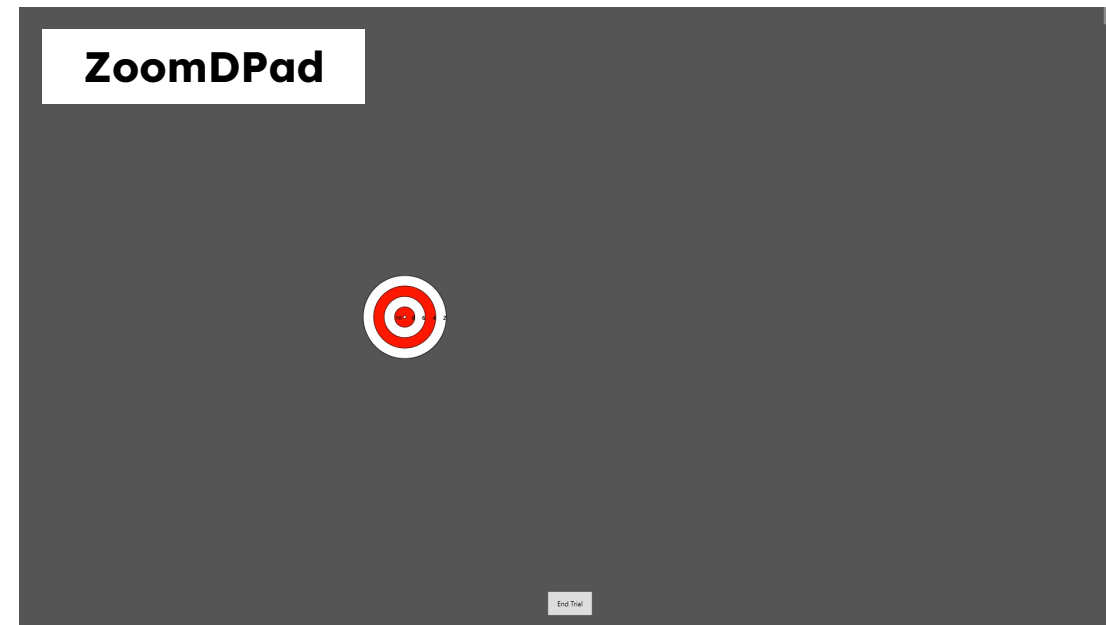
- Under drifted condition, ZoomPursuit higher than ZoomDPad

Usability (UMUX-Lite):

- No significant differences between techniques

Main findings from interviews

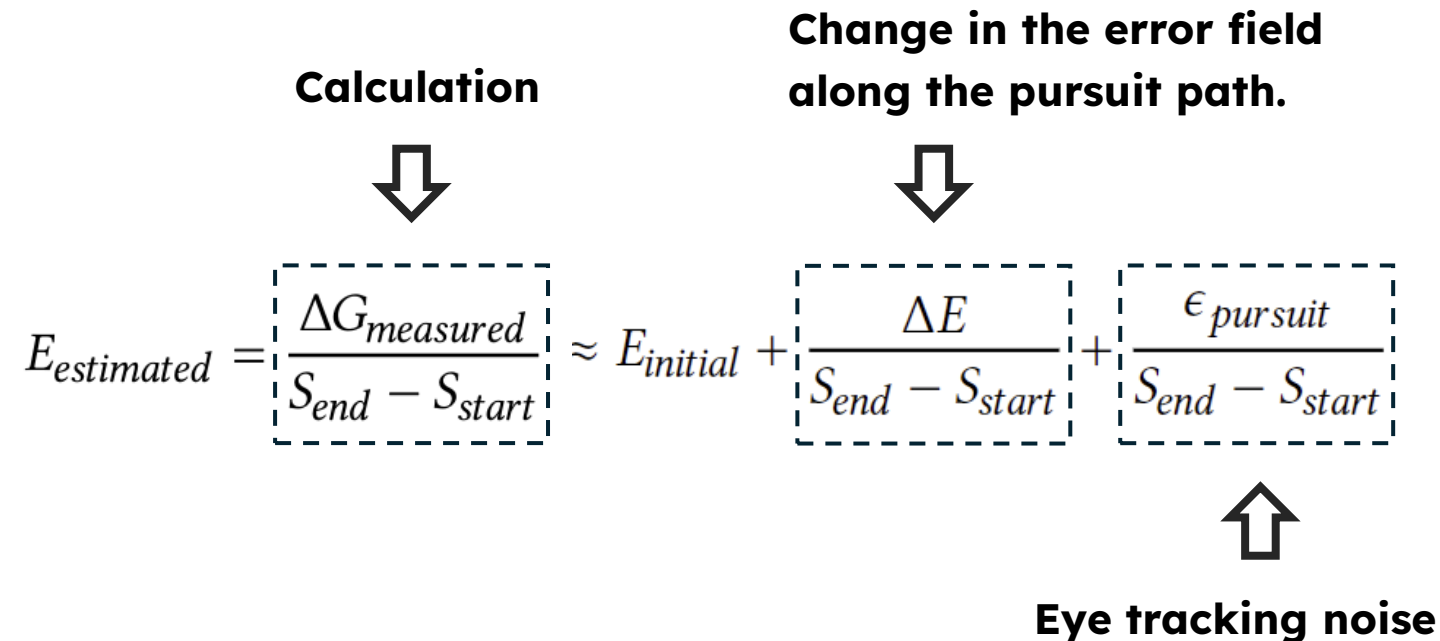
- The ZoomPursuit's auto correction is a black box
- Participants want a sense of control.
 - ZoomDPad: manual gaze refinement;
 - ZoomCenter: overshoot;
 - ZoomPursuit: none



Motivation of a Follow-up Study

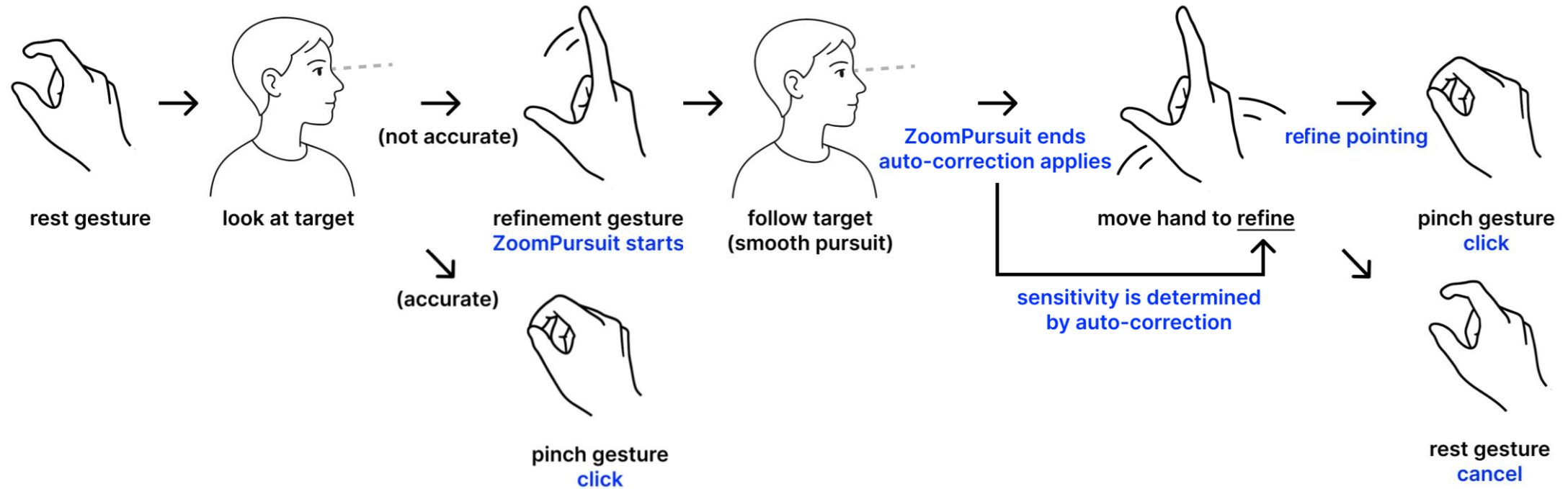
Overall design implications:

- Calibrated but drifted: ZoomPursuit alone is sufficient ✓
- **Not calibrated: ZoomPursuit + another modality to refine**

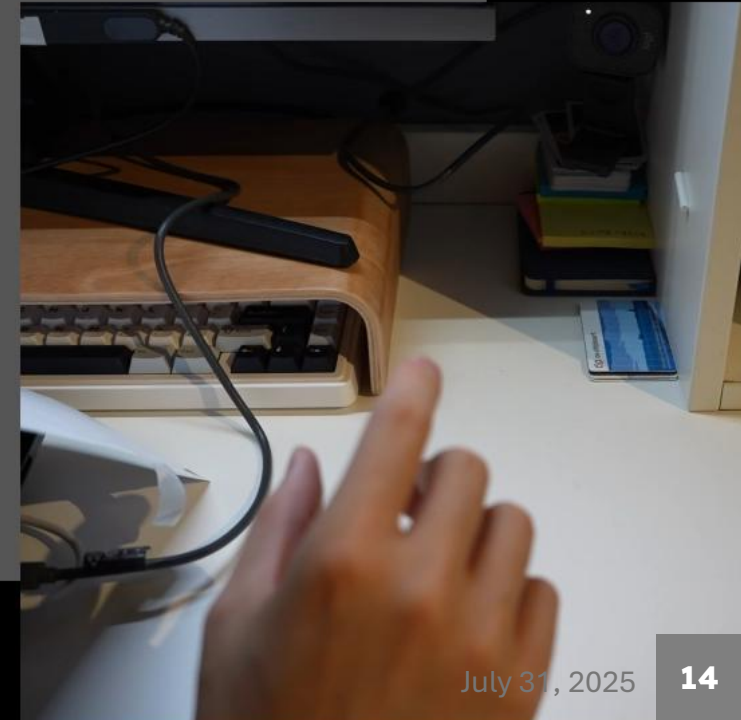


Exploratory Follow-up Study

A Two-stage refinement design based on Gaze + Pinch technique under uncalibrated situation

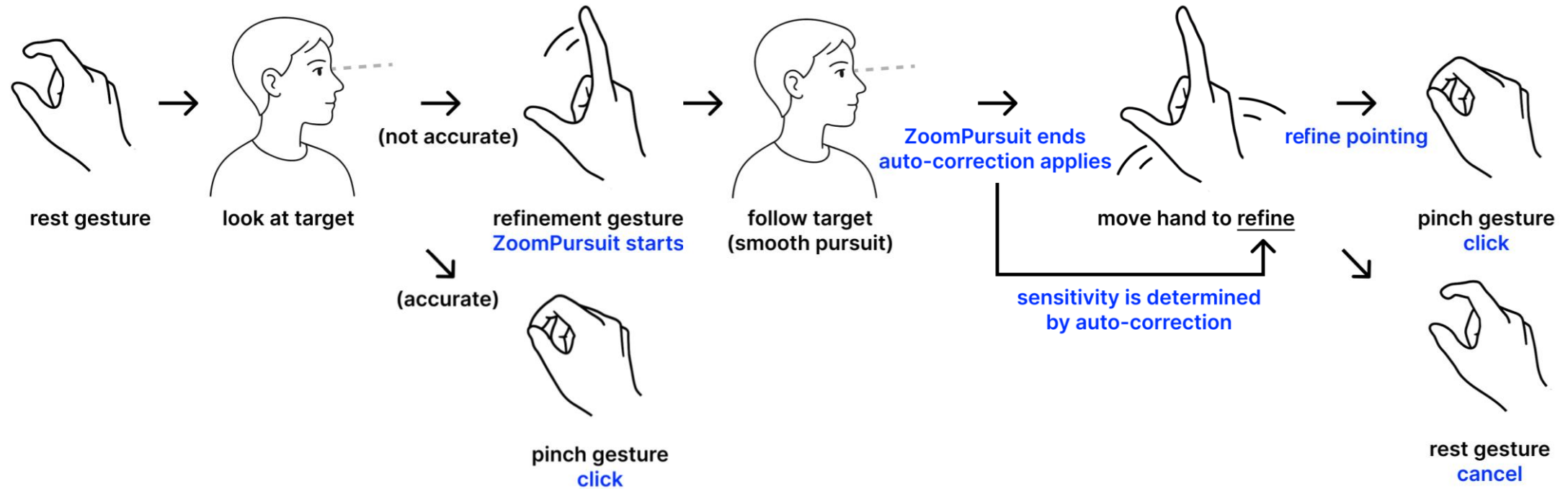


Total Score: 0



Exploratory Follow-up Study

A Two-stage refinement design based on Gaze + Pinch technique under uncalibrated situation



RQ2: How accurate is the ZoomPursuit correction (first stage)?

RQ3: How accurate is the two-stage design?

RQ4: Can the two-stage refinement technique support on-the-fly calibration during interaction?

RQ5: What is the user experience of using the two-stage refinement design?

Human–computer interaction

50 languages

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Human–computer interaction (HCI) is the process through which people operate and engage with computer systems. Research in HCI covers the design and the use of computer technology, which focuses on the interfaces between people (users) and computers. HCI researchers observe the ways humans interact with computers and design technologies that allow humans to interact with computers in novel ways.^[1] These include visual, auditory, and tactile (haptic) feedback systems, which serve as channels for interaction in both traditional interfaces and mobile computing contexts.^[2] A device that allows interaction between human being and a computer is known as a "**human–computer interface**".

As a field of research, human–computer interaction is situated at the intersection of computer science, behavioral sciences, design, media studies, and several other fields of study. The term was popularized by Stuart K. Card, Allen Newell, and Thomas P. Moran in their 1983 book, *The Psychology of Human–Computer Interaction*. The first known use was in 1975 by Carlisle.^[3] The term is intended to convey that, unlike other tools with specific and limited uses, computers have many uses which often involve an open-ended dialogue between the user and the computer. The notion of dialogue likens human–computer interaction to human-to-human interaction: an analogy that is crucial to theoretical considerations in the field.^{[4][5]}

Introduction [edit]

Humans interact with computers in many ways, and the interface between the two is crucial to facilitating this interaction. HCI is also sometimes termed *human–machine interaction* (HMI), *man-machine interaction* (MMI) or *computer-human interaction* (CHI). Desktop applications, web browsers, handheld computers, and computer kiosks make use of the prevalent graphical user interfaces (GUI) of today.^[6] Voice user interfaces (VUIs) are used for speech recognition and synthesizing systems, and the emerging multi-modal and Graphical user interfaces (GUI) allow humans to engage with embodied character agents in a way that cannot be achieved with other interface paradigms.

The Association for Computing Machinery (ACM) defines human–computer interaction as "a discipline that is concerned with the design, evaluation, and implementation of interactive computing systems for human use and with the study of major phenomena surrounding them".^[6] A key aspect of HCI is user satisfaction, also referred to as End-User Computing Satisfaction. It goes on to say:



A computer monitor provides a visual interface between the machine and the user.

Appearance hide

Text

☐ Small☒ Standard☐ Large

Width

☒ Standard☐ Wide

Color (beta)

☐ Automatic☒ Light☐ Dark

Follow-up Study Results

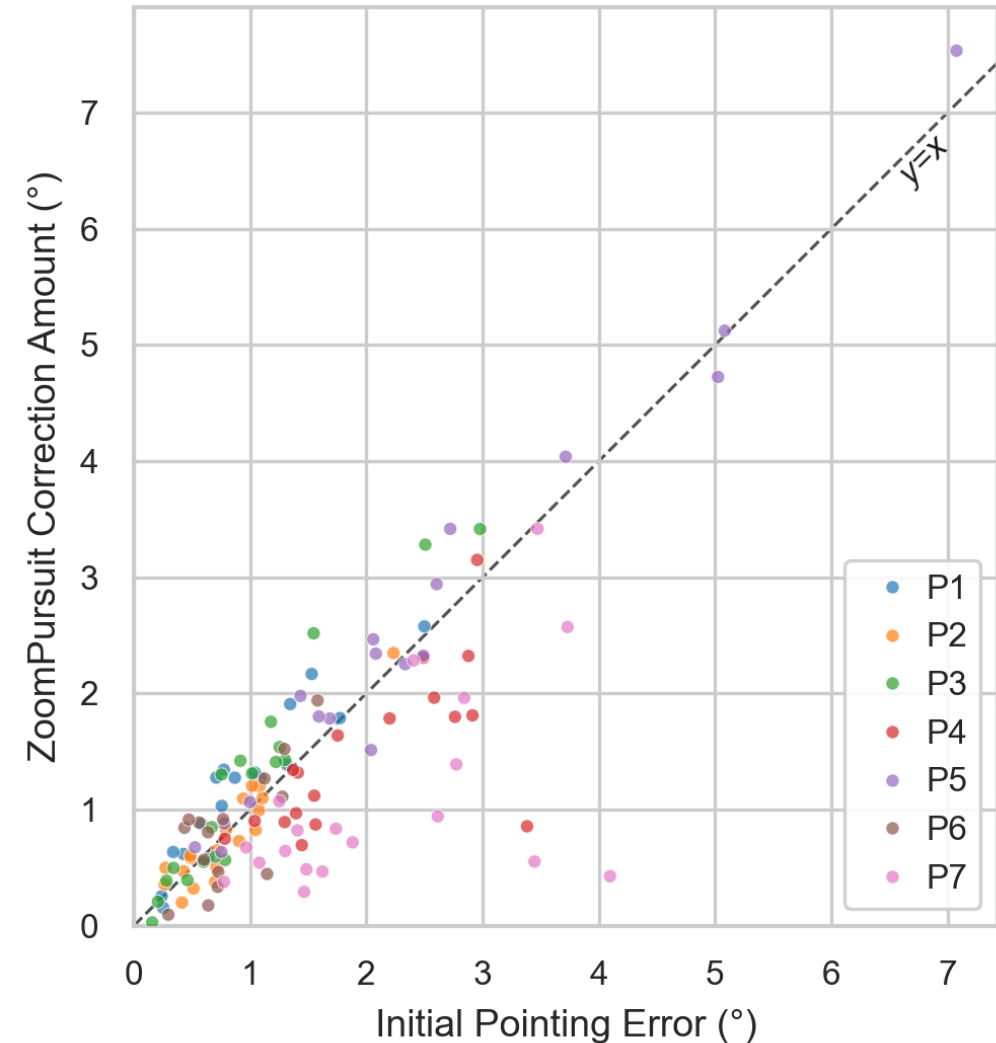
RQ2: *How accurate is the ZoomPursuit correction (first stage)?*

- In average, correct 62% of initial error
- ZoomPursuit correction can reliably predict initial error

RQ3: *How accurate is the two-stage design?*

Accuracy (average of all trials):

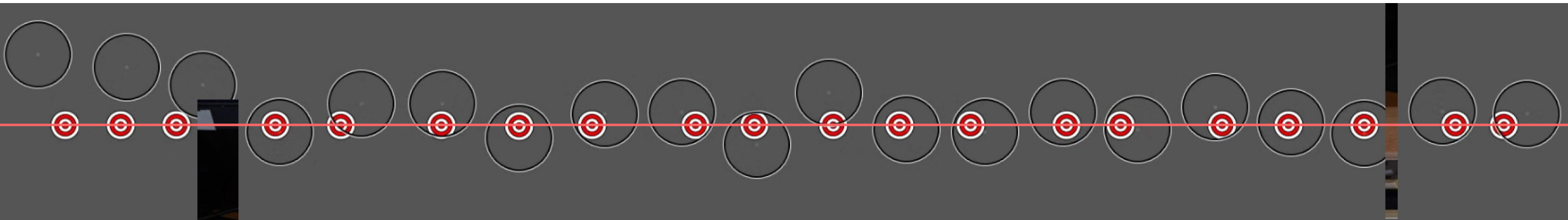
- 1.47°, initial error
- 0.56°, after ZoomPursuit correction (RQ2)
- 0.10° (SD = 0.08), final pointing error (RQ3)



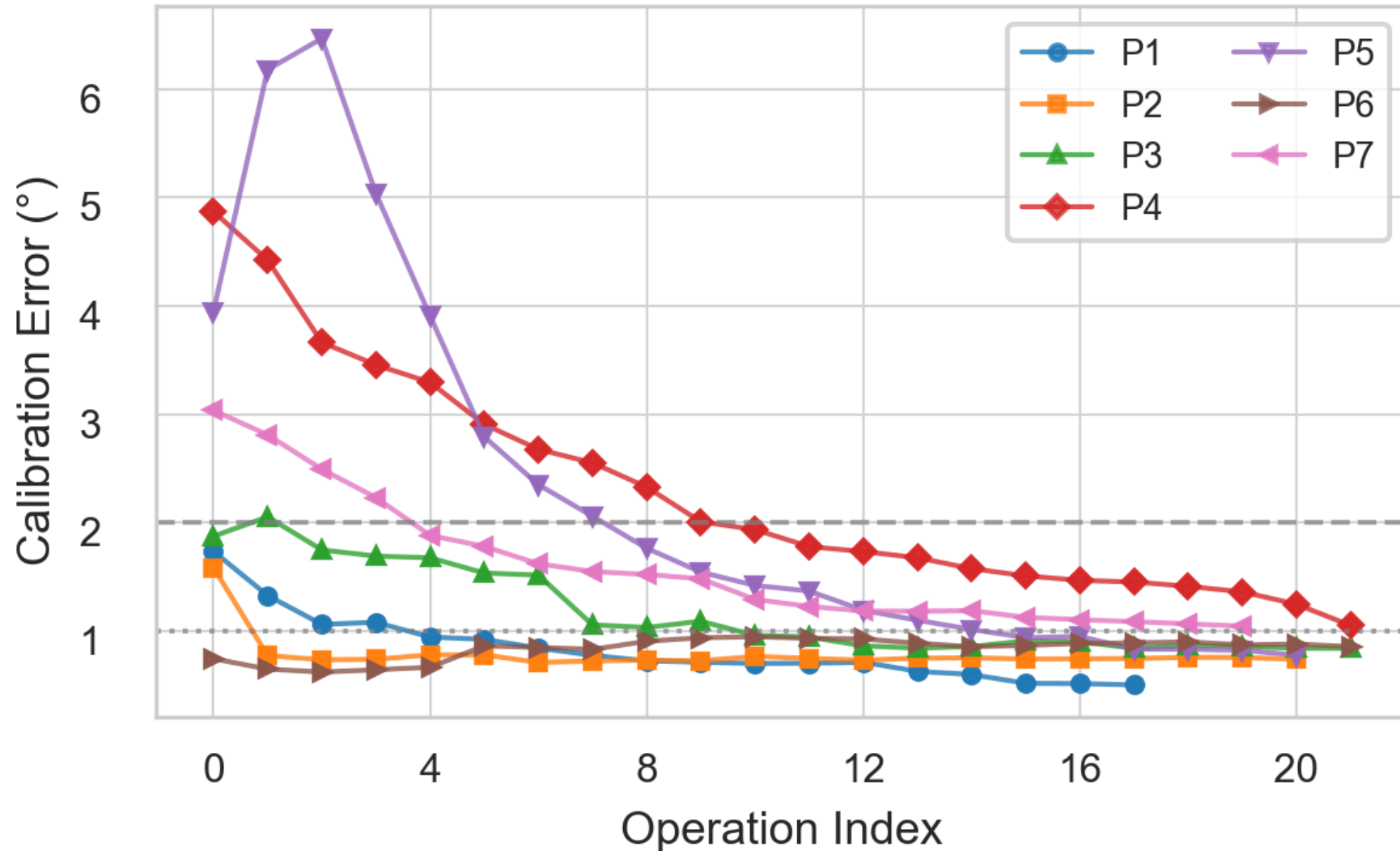
Follow-up Study Results

RQ4: *Can the two-stage refinement technique support on-the-fly calibration during interaction?*

- On the second selection task, after 20 operation
 - From 1.89° (SD=0.83) to **0.84° (SD=0.20)**.
 - Standard 5-point calibration: **0.68° (SD=0.14)**.



Calibration Error in the Selection Task



Follow-up Study Results

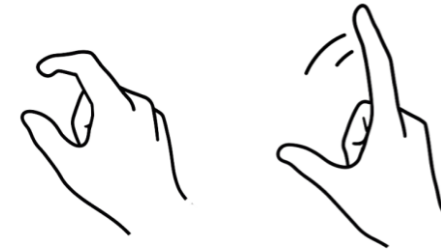
RQ5: *What is the user experience of using the two-stage refinement design?*

Usability: marginal (SUS score: 66.1).

Workload: low (NASA TLX score: 27)

Qualitative findings from interviews

- Ambiguous gesture design reduced usability
- Manageable learning curve of in decoupling eye and hand coordination
- Cognitive Load is higher during WikiGame task



Discussion – Overall Research Question

► Overall research question:

*How can we design a smooth pursuit technique that **retains its benefits** and is **applicable** to general-purpose GUIs?*

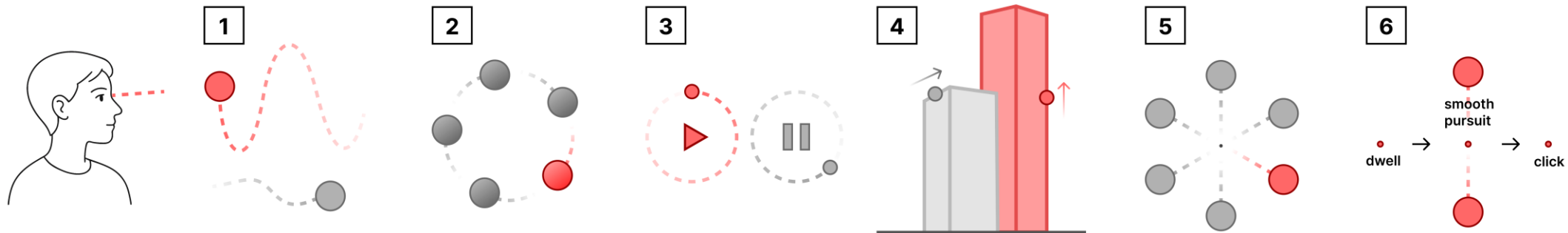
Benefits of smooth pursuit:

- Robust to absolute error / explicit-calibration-free
- Accurate and robust
- Potential for dynamic calibration

ZoomPursuit



**And can work on
general-purpose GUIs!**



Discussion – Possible Applications

Overall design implications:

- Calibrated but drifted: ZoomPursuit alone is enough
- Not calibrated: ZoomPursuit + another modality to refine

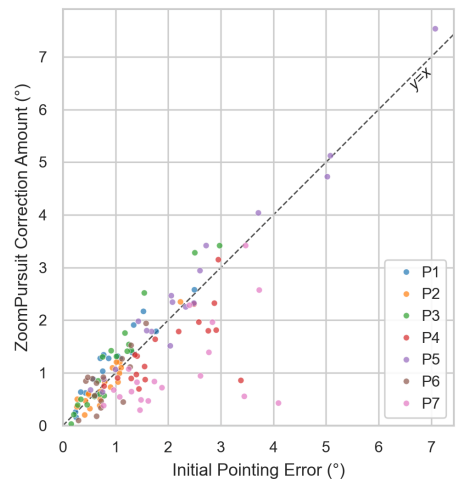
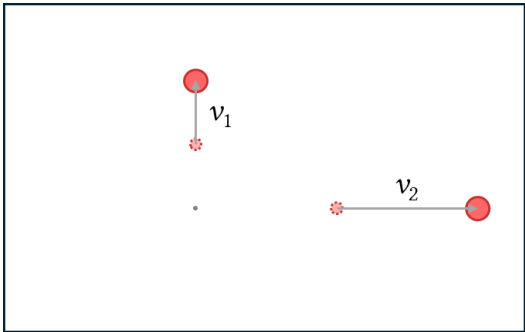
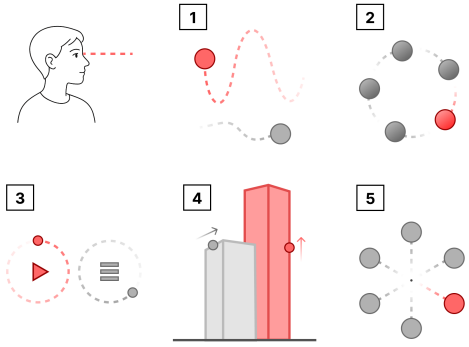
Accessibility: **A promising way to solve calibration drift**

- Posture changes always introduce drift
 - If large, user must calibrate before use.
 - If small, user either tolerate the small drift all the time
- Drift caused by posture changes tend to have same direction
- ZoomPursuit can achieve high accuracy in this case, and dynamically calibrate the error

Reliably predict magnitude of error can be used for:

- Adaptive interaction systems
- Seamlessly downgrade to basic zoom-based technique (perform better under accurate condition)

Thank you!

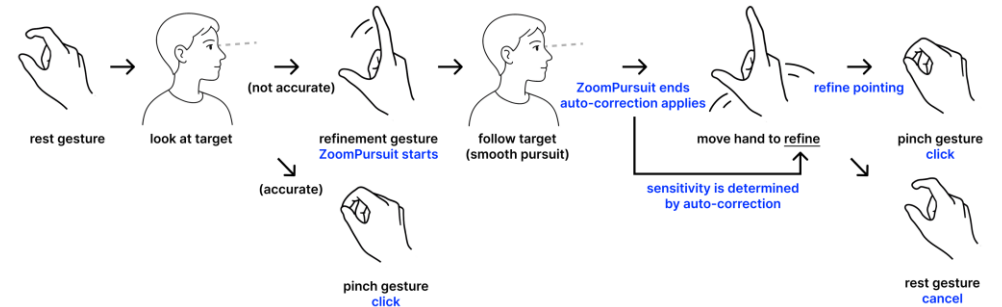
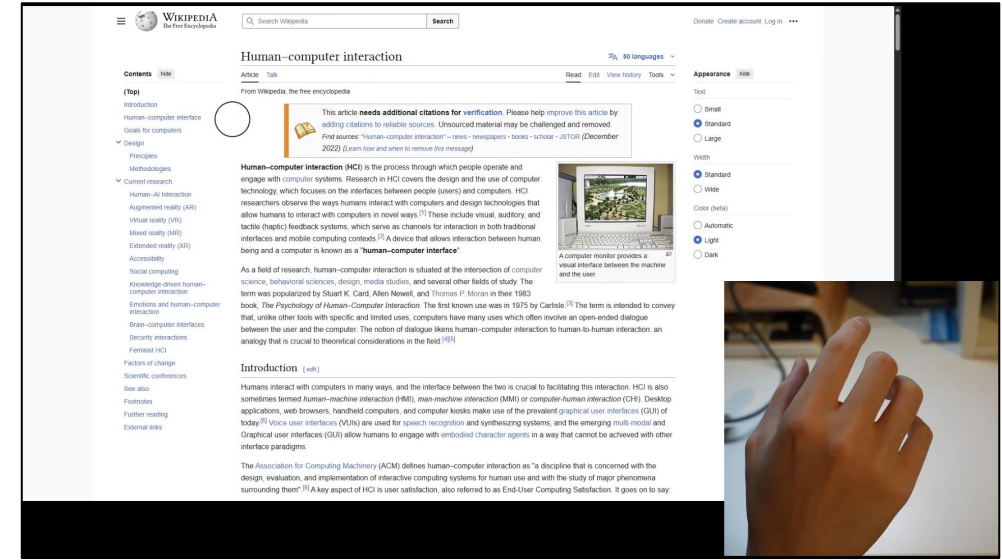
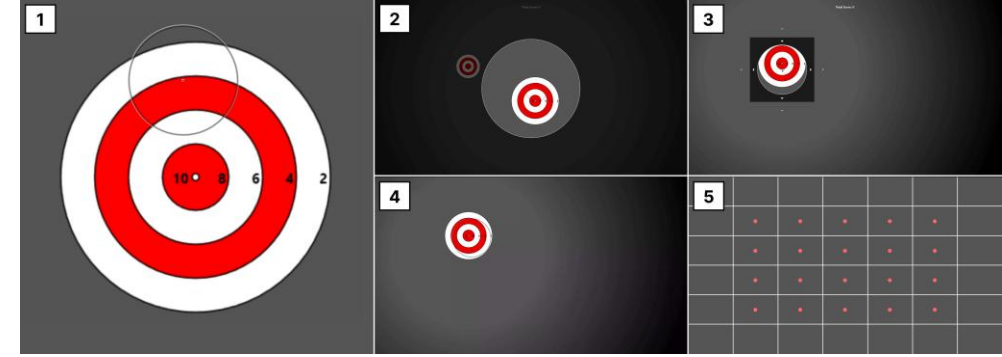
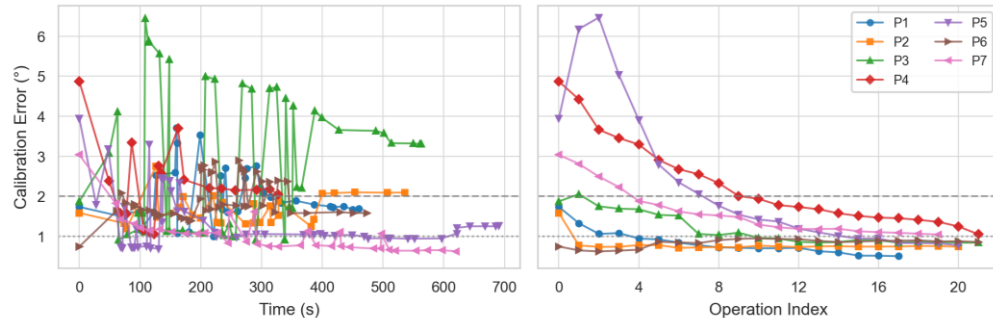
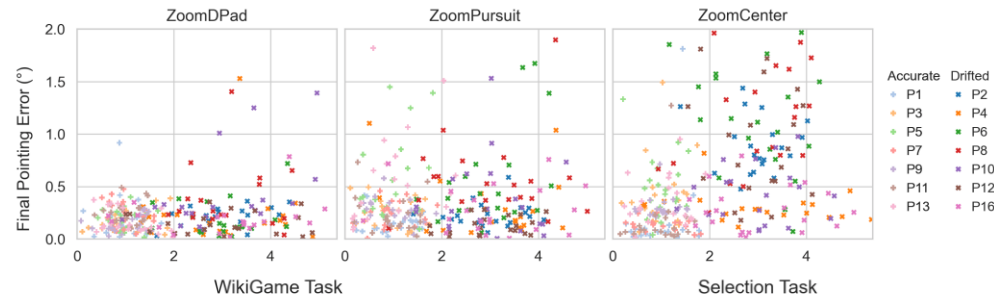
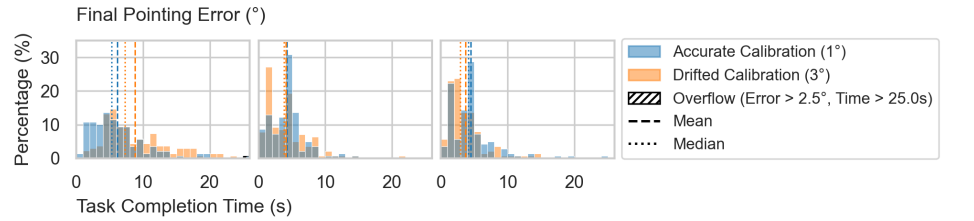
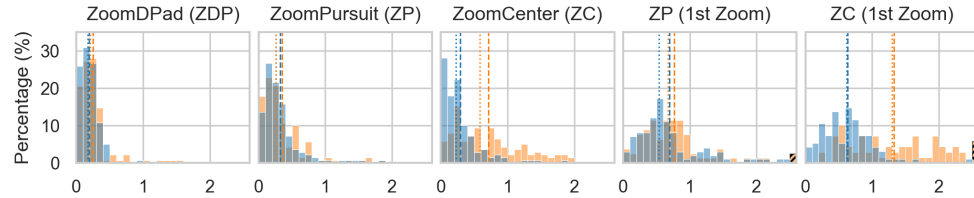


Calculation

$$E_{estimated} = \frac{\Delta G_{measured}}{S_{end} - S_{start}} \approx E_{initial} + \frac{\Delta E}{S_{end} - S_{start}} + \frac{\epsilon_{pursuit}}{S_{end} - S_{start}}$$

Change in the error field along the pursuit path.

Eye tracking noise



Discussion – Possible Applications

Reliably predict magnitude of error can be used for Adaptive interaction systems

Webcam-based eye tracking systems: potential performance improvement

ZoomPursuit for rough pointing and other modality for selection

Discussion

Limitations:

- Non-diverse user sample (students and researchers).
- Gesture design in Study 2: ambiguity between gestures causes mis-activations.
- Small sample size in Study 2 (N=7): only proof-of-concept

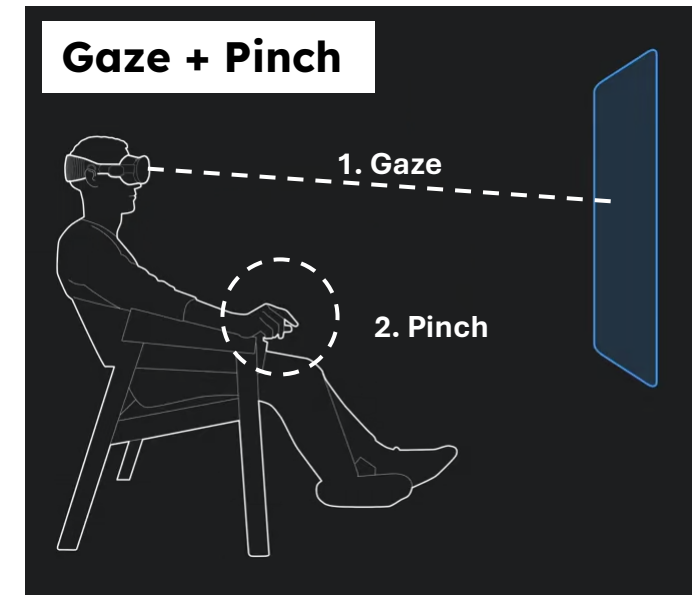
Two User Studies

1. Baseline Evaluation

RQ1: *How do ZoomPursuit and two baseline techniques compare in **performance** and **user experience** under **accurate** and **drifted calibration conditions**?*

2. Exploratory Follow-up Study

- A Two-stage refinement design based on Gaze + Pinch
- Starts from uncalibrated condition
- Dynamic on-the-fly calibration
- 4 RQs



Calculation & Performance Analysis

(Check thesis section 3 for details)

Use relative changes of gaze

- Explicit-calibration-free interaction

Calibration drift is often consistent

- Robust to calibration drift

Error field is often locally stable

- Accurate even uncalibrated

Calculation

↓

$$E_{estimated} = \frac{\Delta G_{measured}}{S_{end} - S_{start}} \approx E_{initial} + \frac{\Delta E}{S_{end} - S_{start}} + \frac{\epsilon_{pursuit}}{S_{end} - S_{start}}$$

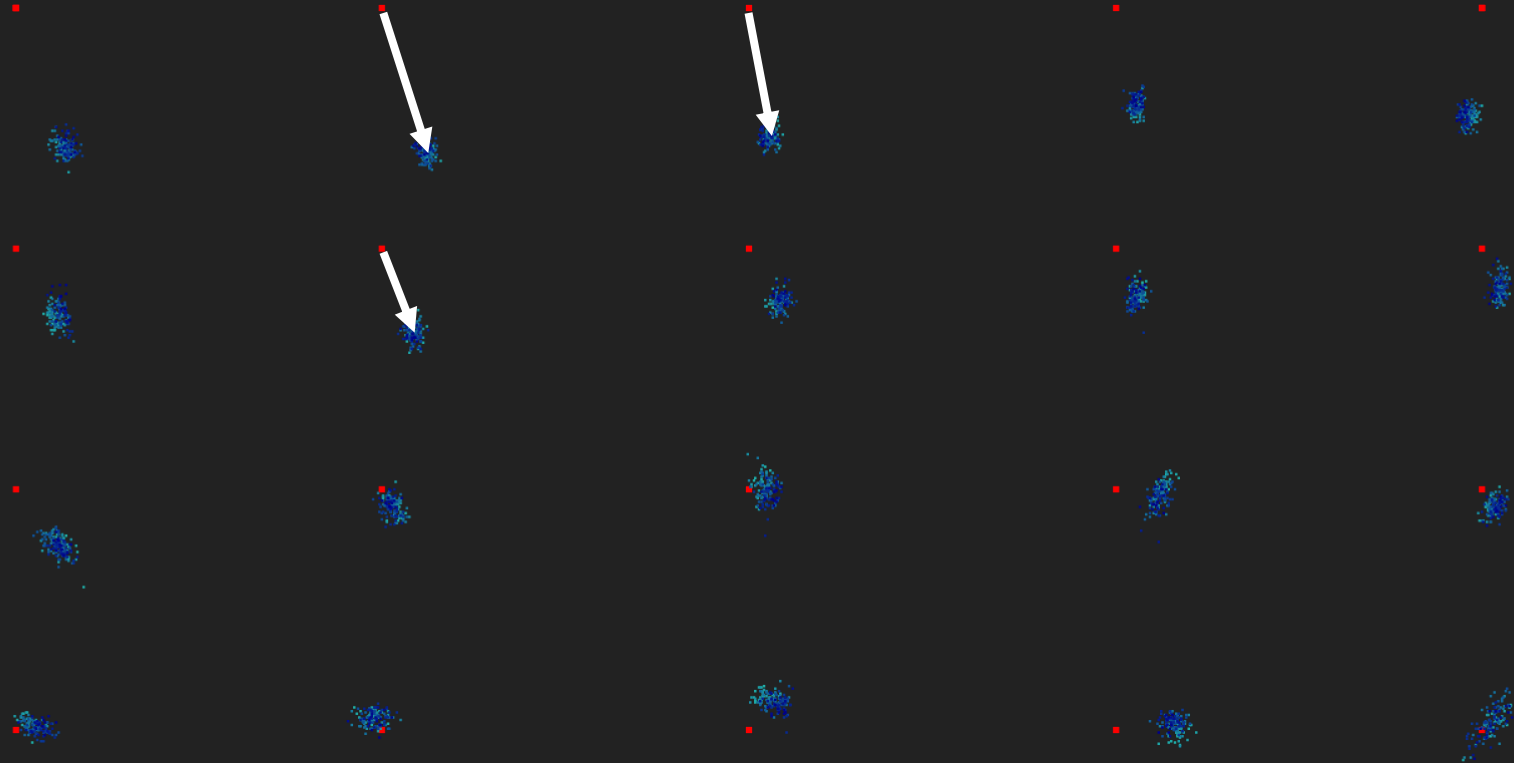
Change in the error field along the pursuit path.

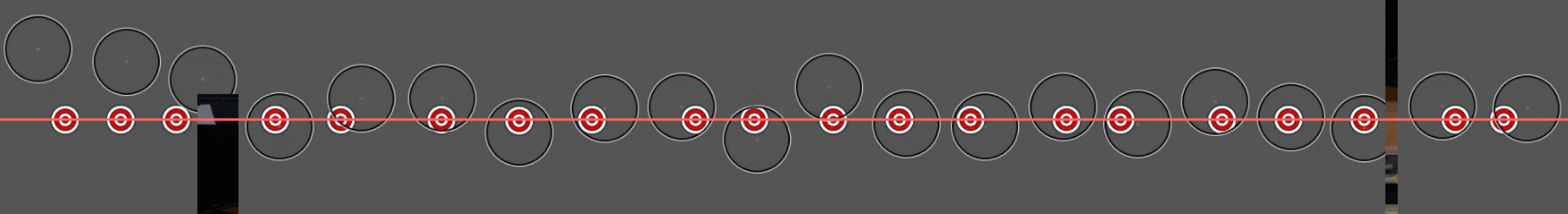
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Eye tracking noise

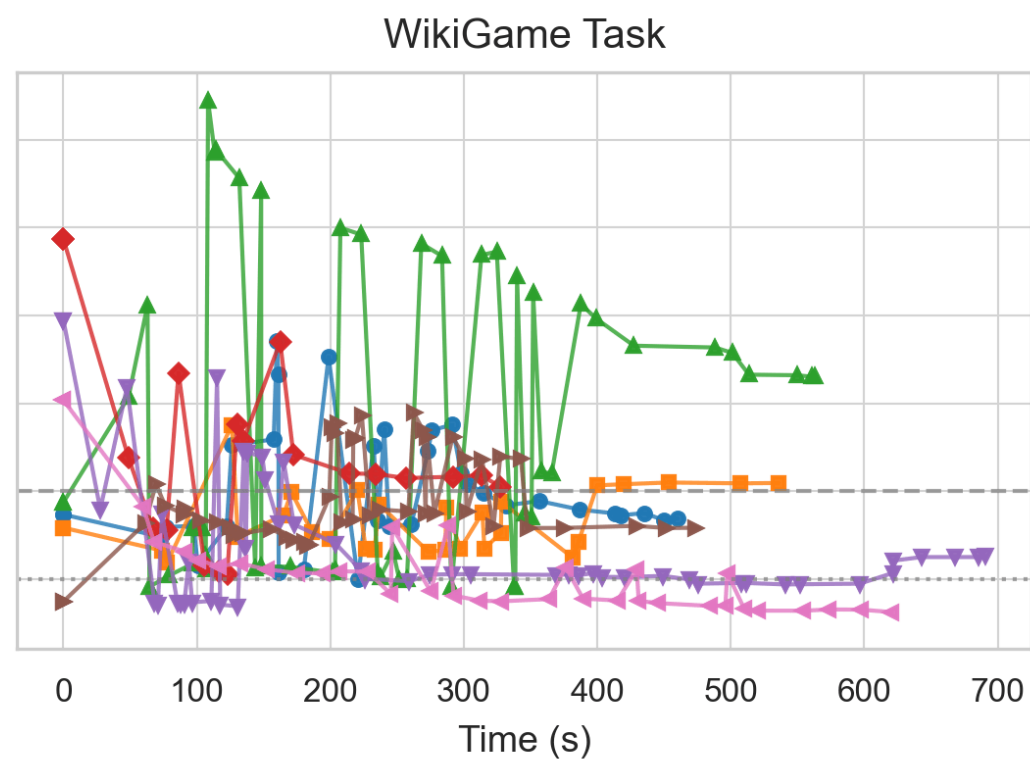
↑

Error Field Visualization





Operation Index 1 → 20



Baseline Evaluation – Preference

Preference:

Under accurate condition: ZoomCenter (4) > ZoomDPad (3) > **ZoomPursuit** (1)

Under drifted condition: **ZoomPursuit** (4) > ZoomDPad (3) > ZoomCenter (1)



ZoomCenter technique adapted from **TD Control** (Commercial accessibility software)



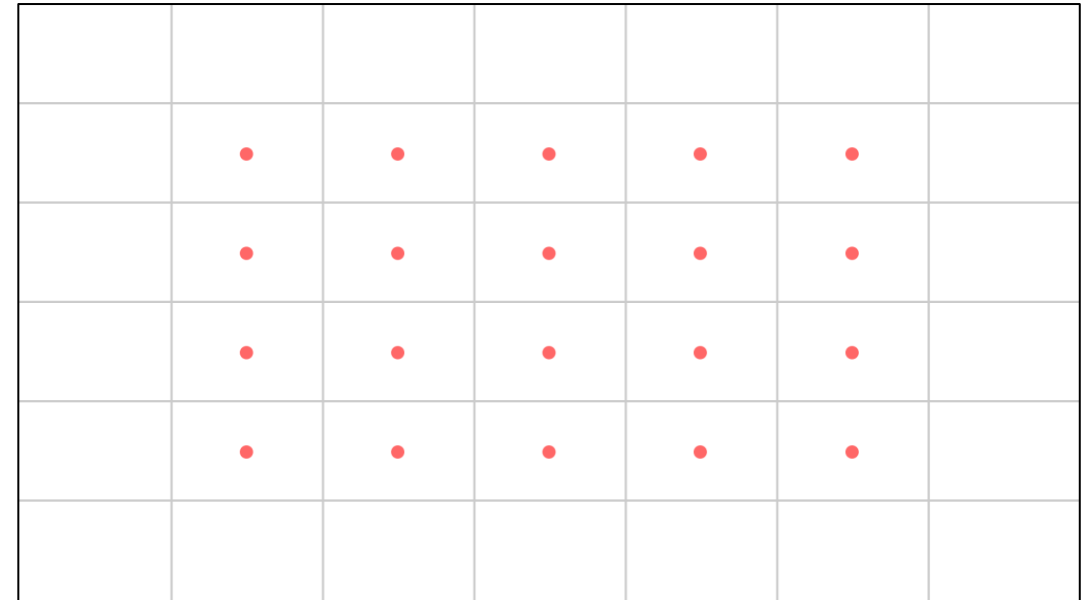
ZoomDPad technique adapted from **Windows Eye Control** (Windows accessibility feature)

Main Procedure of Baseline Evaluation

For each technique:

- 20-dot accuracy sampling
- Apply error factor for assigned calibration level
- 5 practices
- 20 selection tasks (same layout as accuracy sampling dots)
- Post task questionnaire

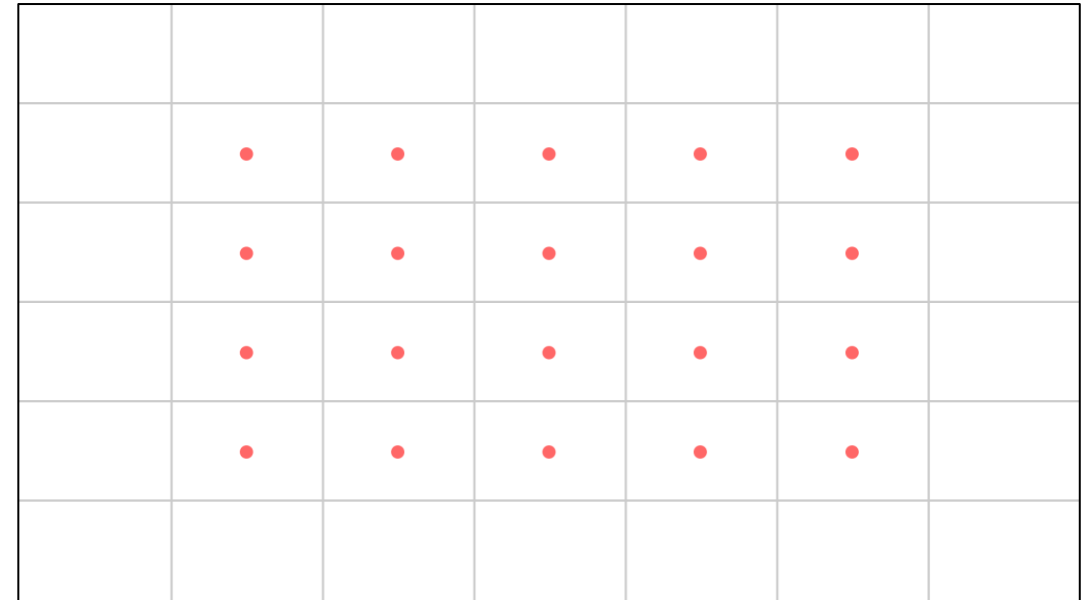
In the end: semi-structure interview



Accuracy sampling dot & target layout (5 × 4)

Main Procedure of Follow-up Study

- 20-dot accuracy sampling
- 10-minute WikiGame task
- SUS questionnaire
- 20-target selection task (same layout as accuracy sampling dots)
- NASA-TLX questionnaire
- Tobii standard 5-point calibration
- 20-dot accuracy sampling
- Semi-structure interview



Accuracy sampling dot & target layout (5 × 4)

Results 1 – Accuracy & Speed

Using Linear Mixed Models (LMMs)

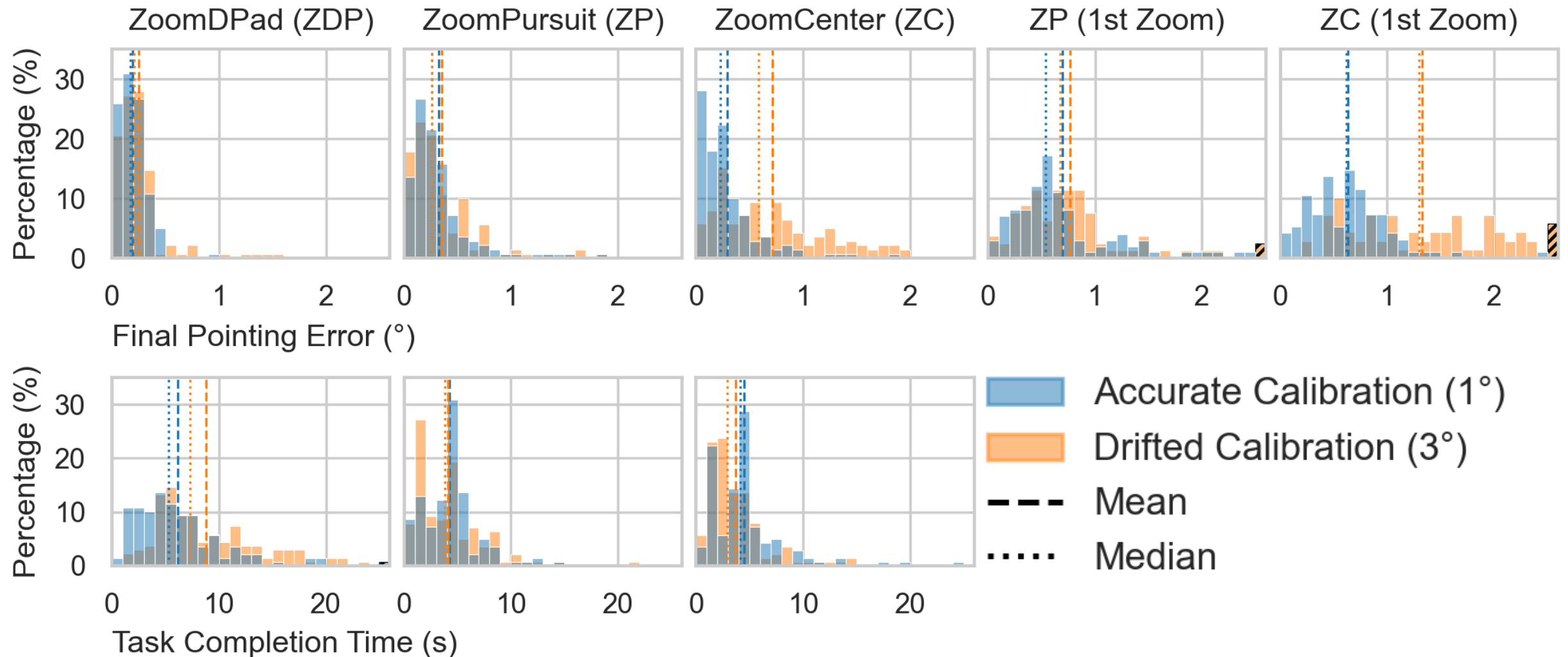
Accuracy (Pointing Error)

- **Under drifted calibration:**
ZoomPursuit and ZoomDPad > ZoomCenter
- **Under accurate calibration:**
All accurate, but
ZoomDPad > ZoomPursuit (small β)

Speed (Task Time)

- **ZoomDPad** is slowest; under drift **90% longer than ZoomPursuit**
- ZoomPursuit $\approx$ ZoomCenter across both calibrations

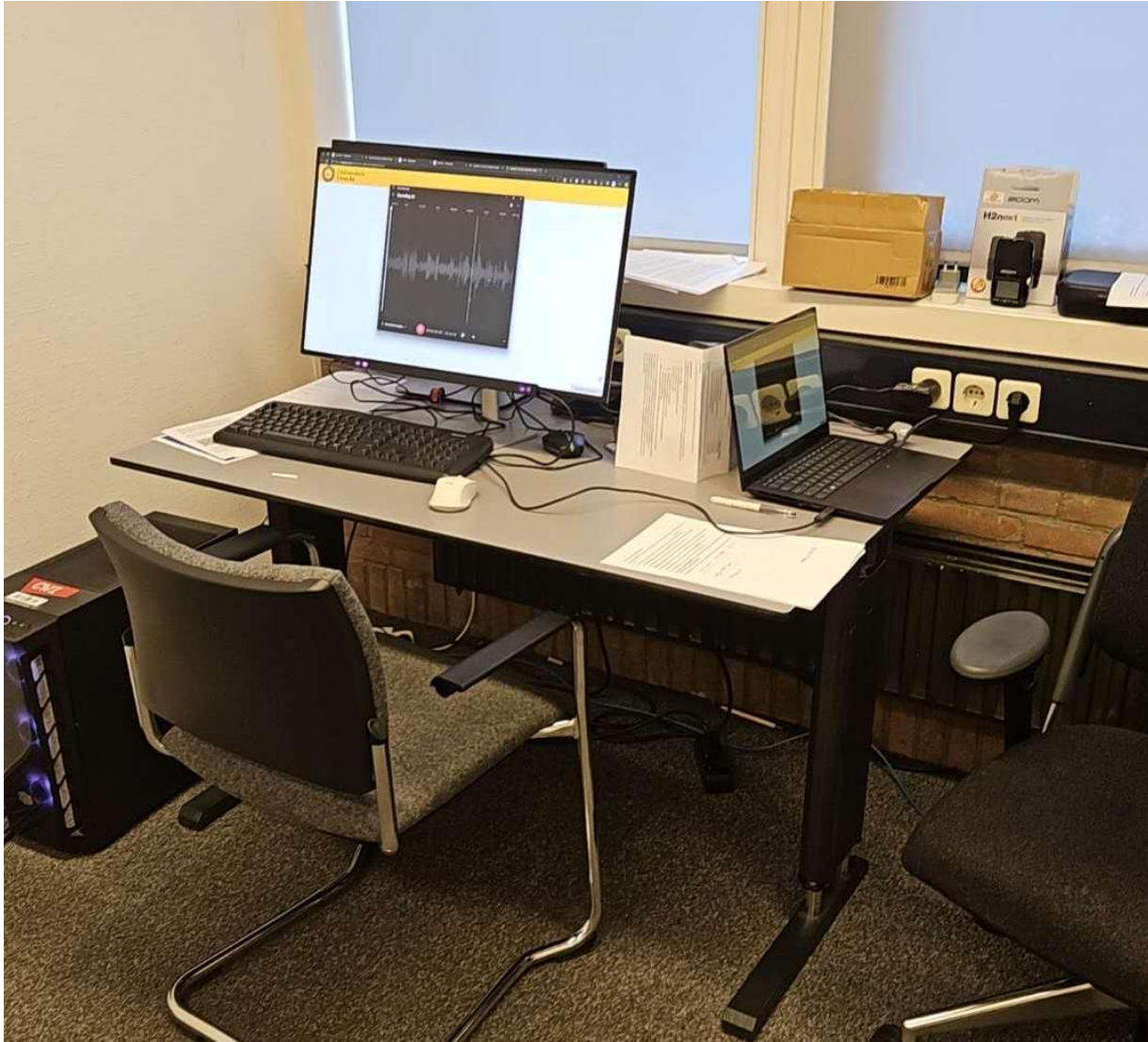
Results 1 – Accuracy & Speed



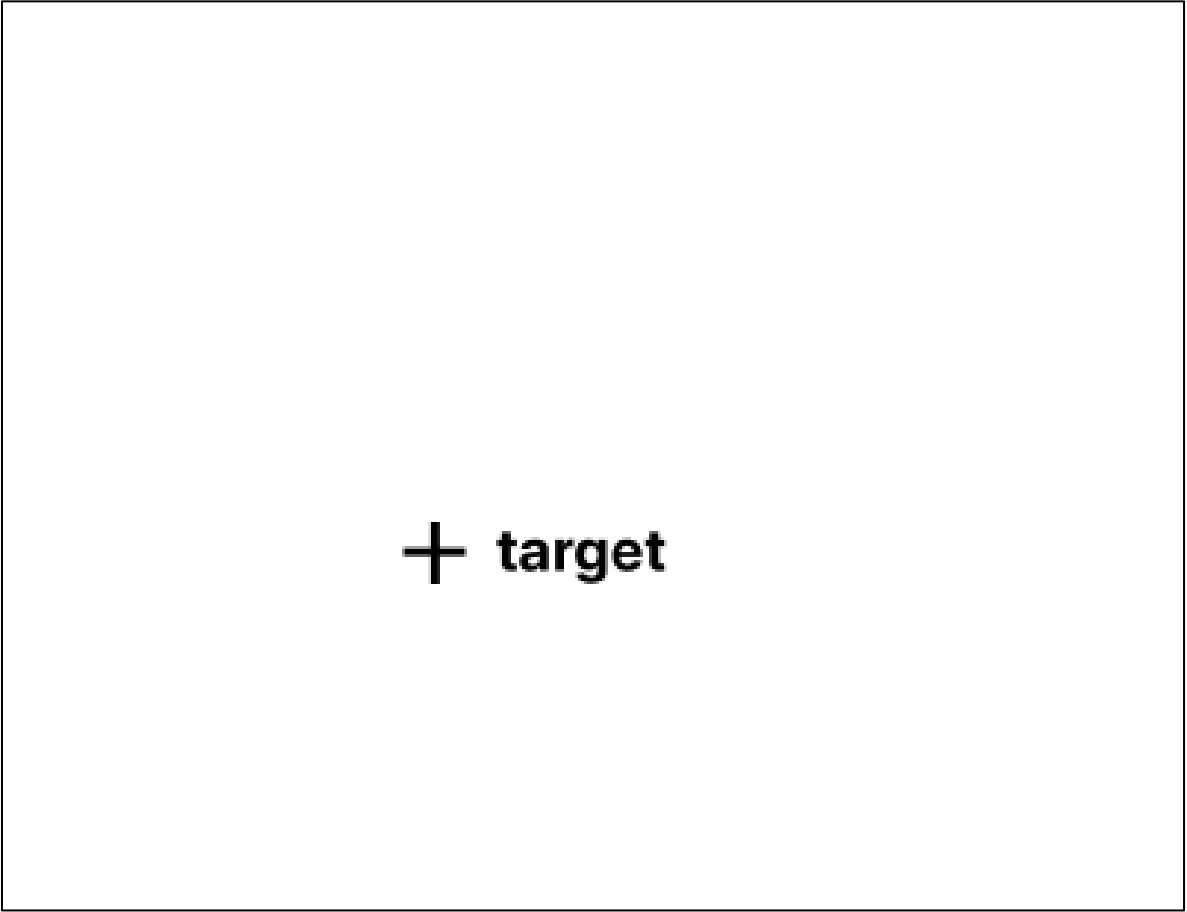
Baseline Evaluation Study Setup



Follow-up Study Setup

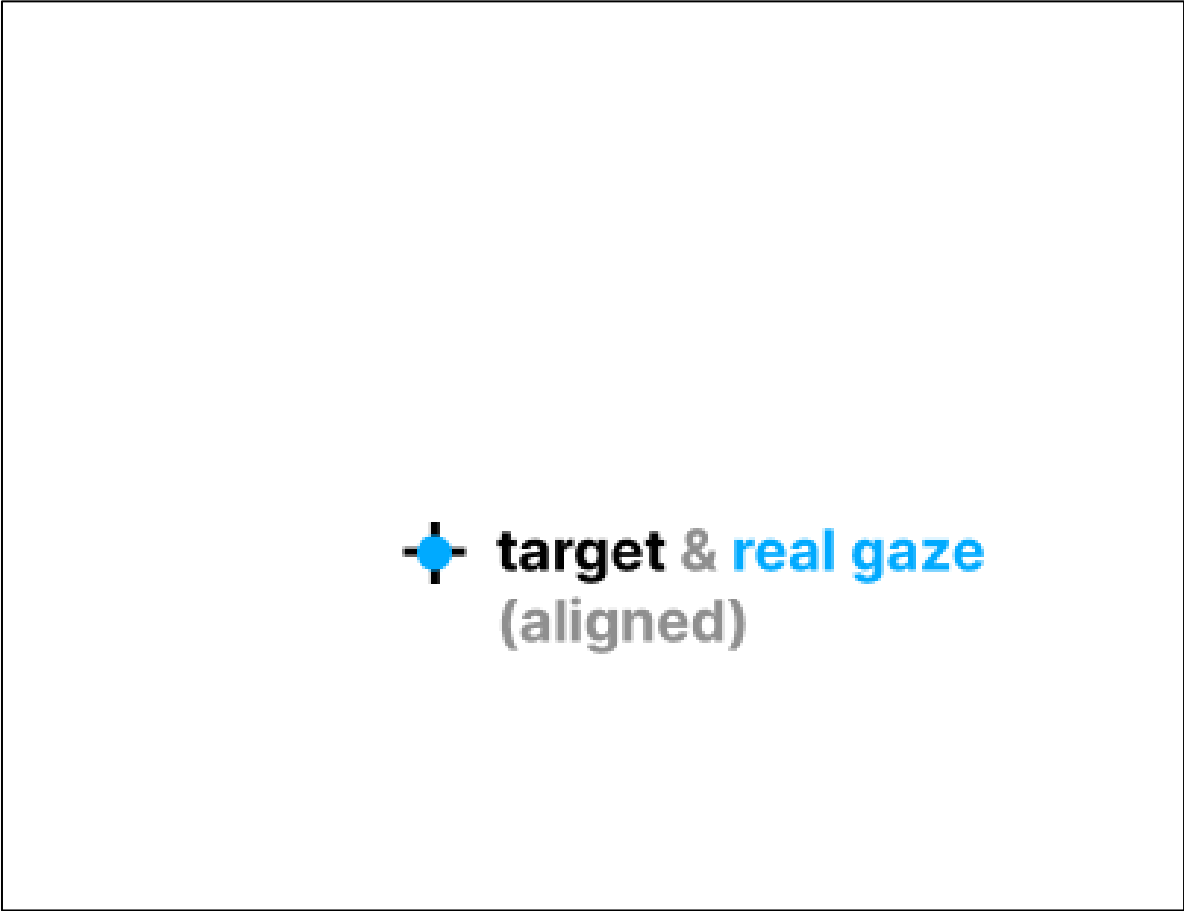


The Zoom Pursuit Technique



+ target

The Zoom Pursuit Technique



 **target & real gaze**
(aligned)

The Zoom Pursuit Technique

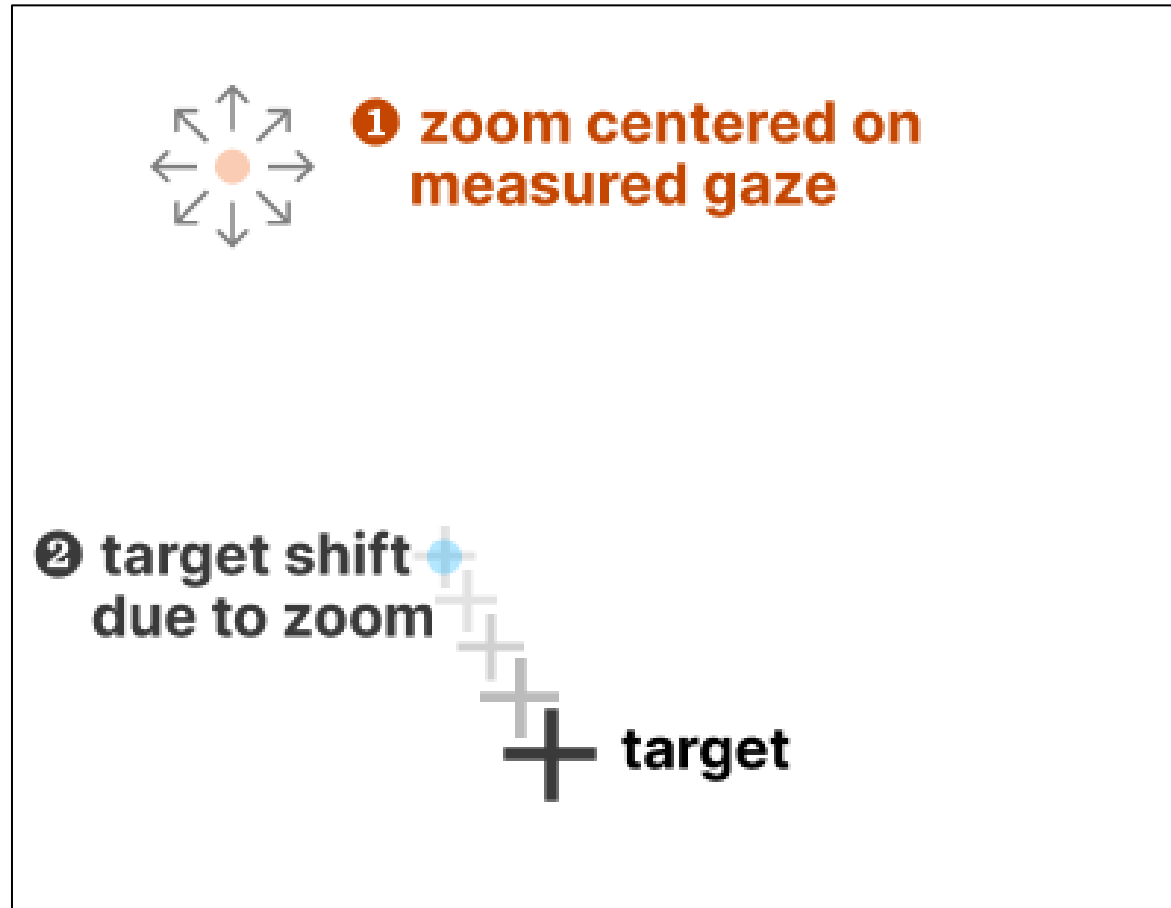
- measured gaze due to tracking error

- ✚ target & real gaze
(aligned)

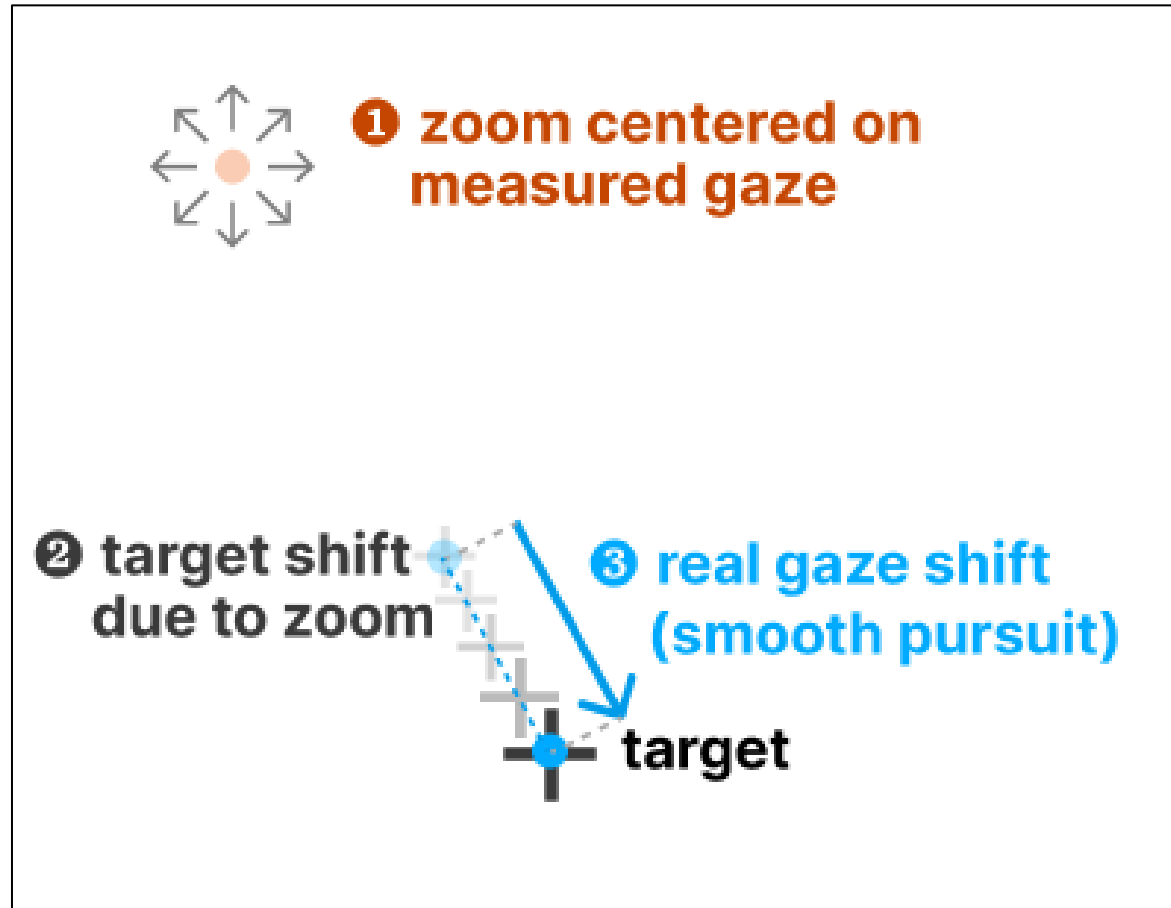
The Zoom Pursuit Technique



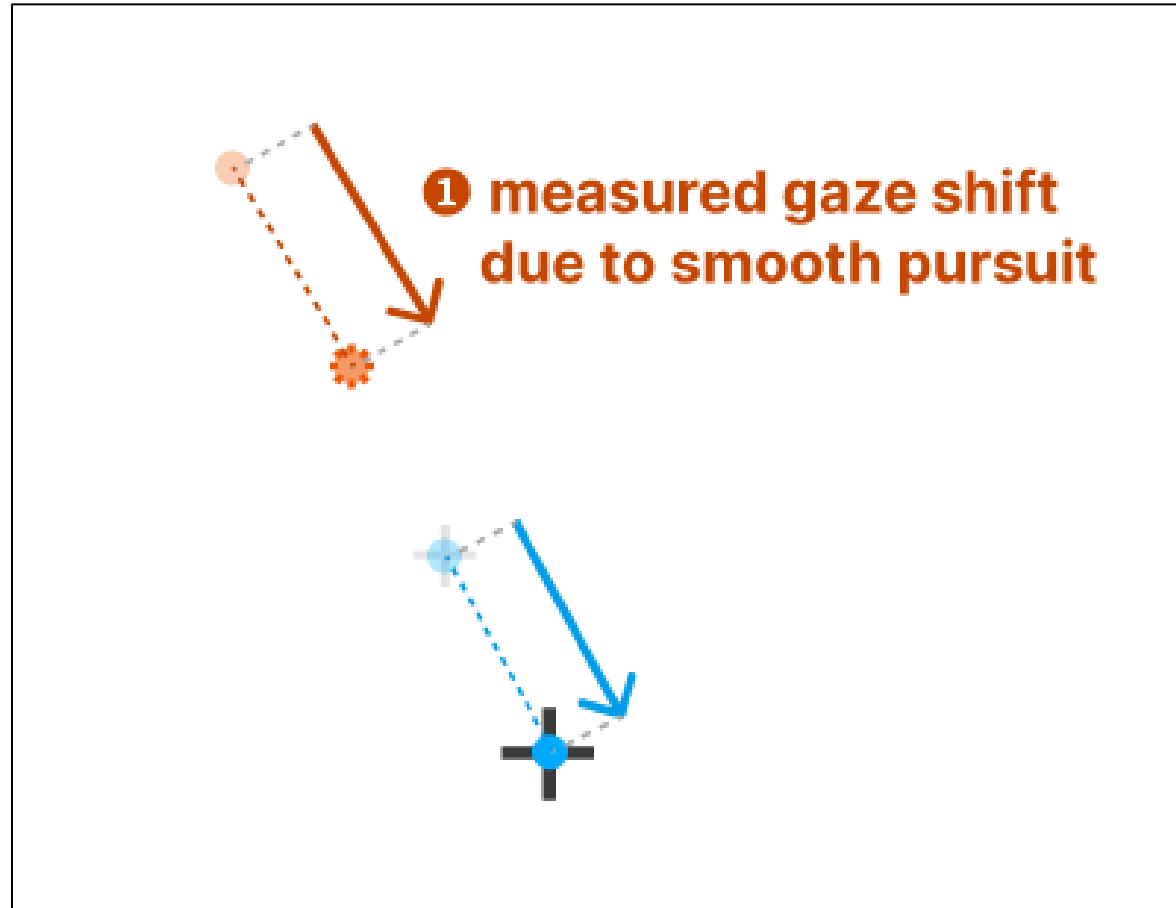
The Zoom Pursuit Technique



The Zoom Pursuit Technique



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The Zoom Pursuit Technique

